AI-Based Preventive Health Risk Assessment Using Lifestyle and Medical Indicators

Dhuruv Kumar Gupta

Abstract

Most people don't think about their health until something goes wrong. By then, it's often too late for simple fixes. This project is about changing that — using AI to catch warning signs early, before they turn into serious problems.

The idea is simple: instead of waiting for a doctor's appointment, a person can enter some basic information about their lifestyle and health things like their weight, blood pressure, sleep habits, and activity level and the system will tell them whether they might be at risk for conditions like diabetes, hypertension, or heart disease.

This research builds a working, AI powered health risk assessment tool using machine learning. The goal isn't to replace doctors it's to give people a heads up, so they know when to see one.

Introduction

Think about how many people skip regular health checkups not because they don't care, but because it costs time, money, or just feels unnecessary when nothing seems wrong. This is exactly how small, preventable conditions grow into serious illnesses.

Diseases like diabetes, high blood pressure, obesity, and heart disease usually develop quietly over years. By the time symptoms show up, the damage has already been done. The frustrating part is that most of these conditions could be detected and managed much earlier if only people had a way to check.

This is where AI comes in. Machine learning models can look at a person's health data and pick up patterns that might point to future health problems patterns that aren't always obvious to the human eye. This project builds a system that does exactly that:

it takes everyday health information and uses it to give a simple, clear risk report.

Problem Statement

Here's the core issue: most healthcare systems are built to treat people who are already sick, not to help healthy people stay that way.

Regular health monitoring is expensive and time consuming. Many people, especially in smaller cities or rural areas, don't have easy access to doctors for routine checkups. And even when they do, a single consultation gives only a snapshot not an ongoing picture of someone's health.

There's a real gap here between the data people already have access to (weight, sleep, activity, blood sugar readings etc.) and the kind of actionable insight that could actually change their behavior. This project tries to close that gap using machine learning.

Objectives

- Build a working AI system that can predict health risks based on lifestyle and medical inputs.
- Train and compare multiple machine learning models to find the most accurate one.
- Create a simple, easy-to-use interface so that anyone not just tech savvy users can interact with it.
- Explore how AI can practically support preventive healthcare, not just in theory but in a real application.
- Help users understand their health better and encourage them to take early action.

Proposed Methodology

The system will work in a few straightforward steps:

Step 1 — Collecting Health Data: Users enter their basic health details age, BMI, blood pressure, glucose level, heart rate, how much they sleep, how active they are, and a few lifestyle habits (smoking, diet, stress, etc.).

Step 2 — Cleaning and Preparing the Data: Before feeding data into the model, it needs to be cleaned missing values handled, numbers normalized, and only the most useful features kept. This is standard preprocessing, but it makes a big difference in how well the model performs.

Step 3 — Training Machine Learning Models: Several algorithms will be tested and compared:

- **Logistic Regression** : simple, fast, and interpretable
- **Decision Trees** :easy to understand and explain
- **Random Forest** :generally more accurate, handles messy data well
- **Support Vector Machine (SVM)**: effective for classification tasks

The best-performing model will be selected and integrated into the application.

Step 4 — Building the Application:

- **Backend:** Python with FastAPI lightweight, fast, and easy to deploy
- **Frontend:** React.js a clean, interactive dashboard where users can enter their data and view results
- **Output:** The system will show a risk level (low / moderate / high) with a short explanation and a suggestion to consult a doctor if needed

Step 5 — Evaluation: The system will be evaluated on accuracy, how well it handles real-world inputs, and whether a regular user can actually navigate it without confusion.

Expected Outcomes

- A fully working health risk assessment web application.
- A trained ML model that can reliably classify health risk from user inputs.
- A user interface that feels approachable and easy to use — not like a medical form.
- Practical evidence that AI can play a meaningful role in preventive healthcare.
- A foundation that can be extended later — more diseases, more data, deeper predictions.

Future Scope

There's a lot of room to grow from here. Once the core system works, a few natural next steps would be:

- Connecting with wearable devices (smartwatches, fitness bands) for real-time data

- Adding deep learning models for more complex pattern recognition
- Expanding to cover specific diseases like cancer risk, kidney disease, or mental health
- Making it multilingual so it can reach a broader population
- Deploying it as a cloud-based service for accessibility at scale

Conclusion

Healthcare doesn't always have to start at a clinic. Sometimes it can start with a simple question: *"Am I at risk?"*

This project tries to answer that question using data people already have, models that are interpretable, and an interface that doesn't require any medical knowledge to use. The goal is to make preventive healthcare a little more accessible, a little more personal, and a little more human.

It won't replace a doctor. But it might be the reason someone decides to see one sooner.

References

1. Research articles on AI-based clinical decision support and risk prediction systems
2. Scikit-learn Official Documentation : *scikit-learn.org*
3. FastAPI Official Documentation :*fastapi.tiangolo.com*
4. React.js Official Documentation :*react.dev*